

Mahmoud Youssef

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EDUCATION

El Shorouk Academy

Cairo, Egypt

Bachelor of Science (B.Sc.) in Computer Science

Expected June 2026

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Software Engineering

CS50x — Introduction to Computer Science

Cairo, Egypt

Harvard University (via edX), completed through Information Technology Institute (ITI)

2024

PROJECTS

E-Commerce Backend System | *Spring Boot, MySQL, Redis, Stripe*

[GitHub](#) | [Demo](#)

- Built a 57-endpoint multi-tenant **RESTful API** with tenant-isolated queries, **JWT** refresh rotation, and double-entry accounting; enforced concurrency-safe checkout via pessimistic locking, verified across 1,000 simulated concurrent users with zero data races
- Implemented idempotent **Stripe** webhook handling and COD payments alongside a **Redis** cache-aside catalog and coupon engine, validated by 112 automated tests (**JUnit/Mockito**) including 1,000-user concurrency scenarios

LogiBridge — Delivery Integration Platform | *Spring Boot, PostgreSQL, Redis*

[GitHub](#) | [Demo](#)

- Built a delivery order workflow with an explicit state machine and DB + **Redis** dual-layer idempotency keys, preventing duplicate order processing under network retries
- Implemented pessimistic locking with optimistic **@Version** conflict retries for safe concurrent order updates; secured the **RESTful API** with **JWT** refresh rotation and token-reuse detection

Distributed URL Shortener — SnipLink | *Spring Boot, MySQL, Redis*

[GitHub](#) | [Demo](#)

- Designed a horizontally sharded backend (hash-based routing across 2 MySQL shards) with raw JDBC persistence, eliminating the need for a cross-shard lookup table
- Built **Redis**-backed distributed ID generation, cache-aside redirects, and atomic Lua-script rate limiting with fail-open degradation for continued availability

CalCounter — AI Nutrition Tracker (Team Lead) | *Spring Boot, MySQL*

[GitHub](#) | [Demo](#)

- Led a backend sub-team within a 6+ member cross-functional graduation project, shipping a 30+ endpoint **JWT**/RBAC-secured **REST API** with full **API documentation** via **Swagger/OpenAPI**
- Built an async HuggingFace meal-photo analysis pipeline, cutting processing wait time by ~60%; coordinated **CI/CD** deployment via **Docker** on Render with the frontend team in an Agile/Scrum workflow

TECHNICAL SKILLS

Languages: Java, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, MapStruct, JdbcTemplate, WebSocket/STOMP

Databases: MySQL, PostgreSQL, Redis (caching, sharding, presence, rate limiting)

Security: JWT Authentication, Refresh Token Rotation, RBAC, OAuth2, Keycloak, HMAC-SHA256 Signing, Payment Integrations (Stripe)

Distributed Systems: Idempotency, Circuit Breakers, Outbox Pattern, Sharding, Cache-Aside, Optimistic/Pessimistic Locking

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, Maven, Flyway, Swagger/OpenAPI (API Documentation), Postman

Testing & Methodologies: JUnit 5, Mockito, Testcontainers, Agile/Scrum, SOLID, Design Patterns, Clean Architecture